Imperial College London

# Lecture 5

# Combinational Circuits

Prof Peter YK Cheung

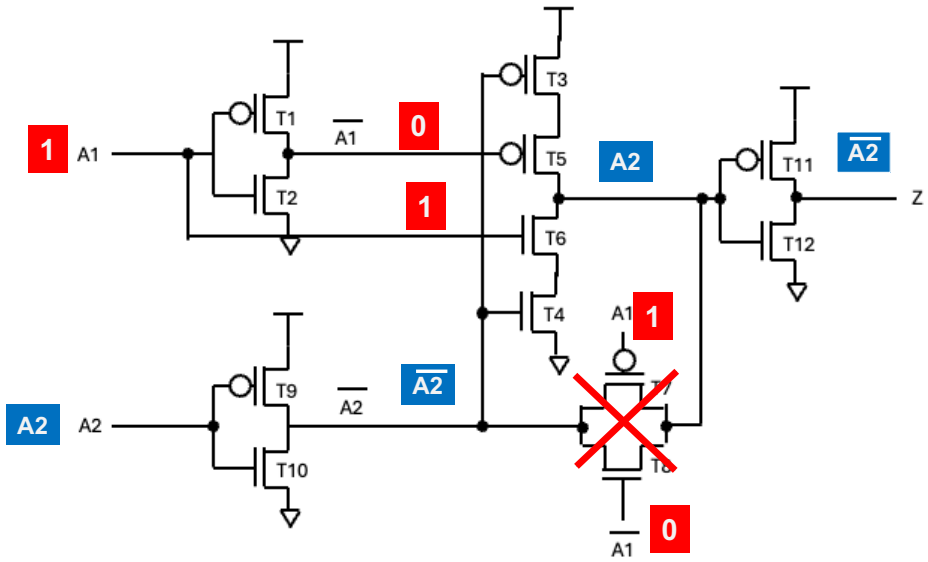
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In this lecture, we will examine the different way that combinational logic gates are implemented in full-custom VLSI circuits.

## XOR gate schematic






This is the solution to Lab 2, Task 2.

When A1 = 1, XOR gate is an inverter to A2.

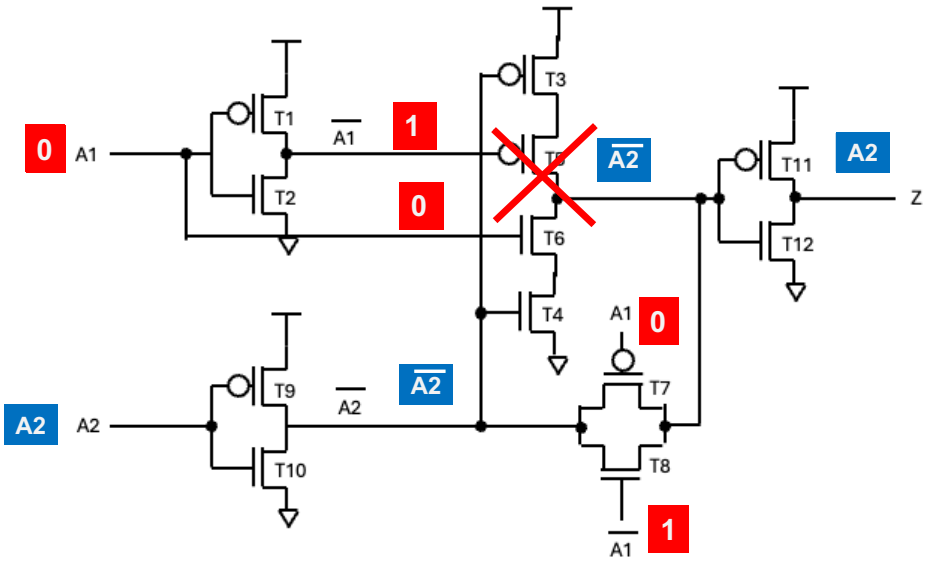
T5 and T6 are ON, T3 and T4 form an inverter for $\overline{A2}$. Therefore, $Z = \overline{A2}$.

When A1 = 0, XOR gate is passing A2 to output without inverting it.

T5 and T6 are OFF. Node Y is NOT driven by T3-T6 circuit.

However, T7 and T8 are ON. They act as a transmission or pass gate, connecting $\overline{A2}$ to node Y. Hence $Z = A2$.

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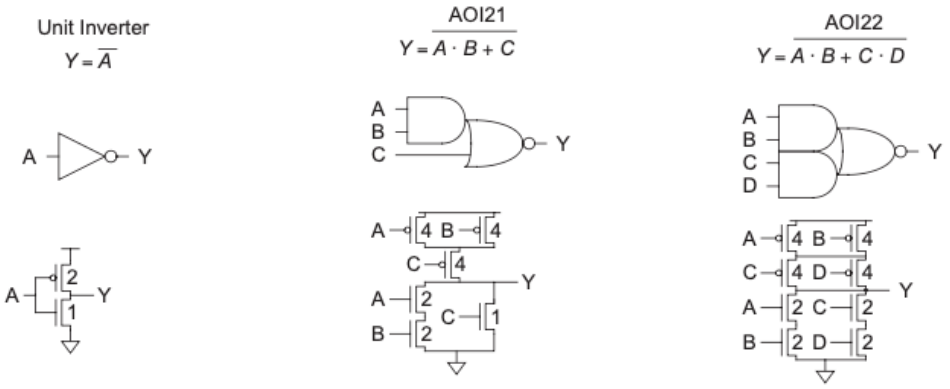
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# Compound Gates




W&H p330



A compound gate is one that implements more than simple AND, NAND, OR, NOR and XOR gate functions. For example, the AOI21 gate is a 2-input AND gate, the output is then Ored with a third input, followed by a inverter (A-O-I).

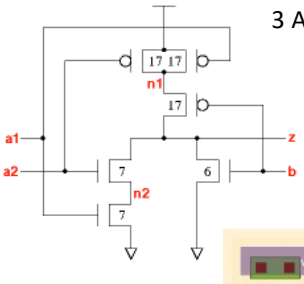
The TMSC standard cell library includes many compound gates.

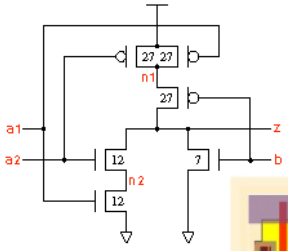
# vsclib Standard Cell Library – aoi21

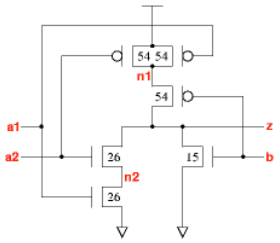


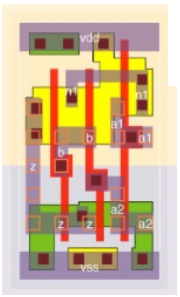

z:
((a1*a2)+b)'

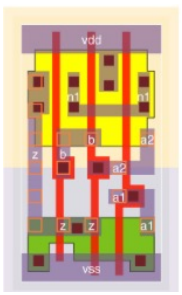
3 AOI31 cells with different drive strengths.

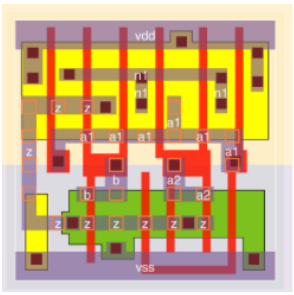



PYKC 15 Oct 2025 EE4 – Digital VLSI Design Lecture 5 Slide 5

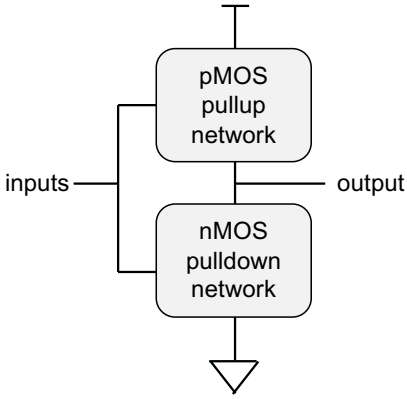
You can find an open-source standard cell library here:

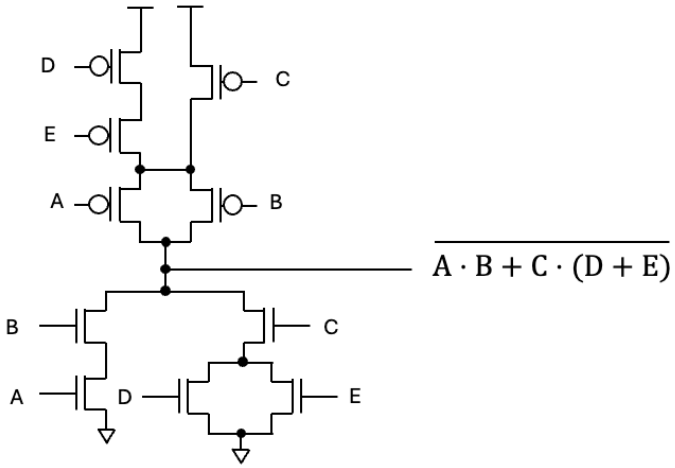
https://www.vlsitechnology.org/html/cells/vsclib013/lib_gif_index.html

This library contains **319** inverting and non-inverting gates, **1** flip-flop and **2** latches. Each cell has a typical timing and input pin capacitance in 0.13µm generic technology, along with its leakage and dynamic power, size, layout, and the transistor schematic.

Transistor sizes are given in lambda, where for this 0.13µm technology, lambda=0.055µm. The cell height is 72 lambda (3.96µm) with power supply rails 8 lambda (0.44µm) wide. The maximum P and N transistor widths before folding are 28λ and 20λ respectively.

# Complex Compound Gate



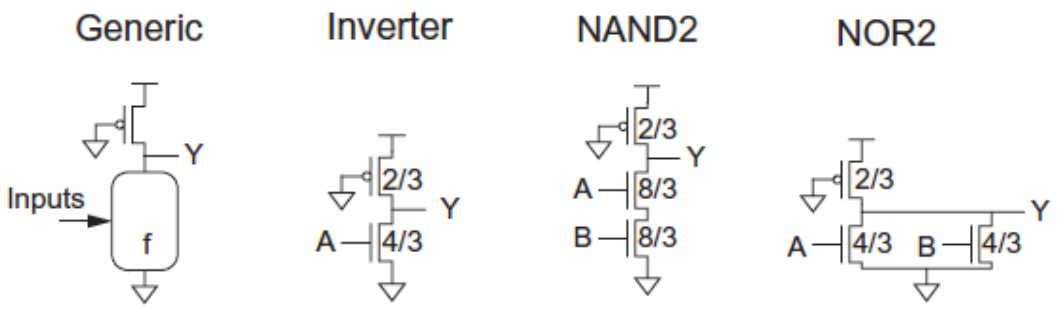






We can form complex combinational circuit function in a complementary tree. The procedure to construct a complementary tree is as follow:-

- Express the Boolean expression in an inverted form
- For the n-transistor tree, working from the inner-most bracket to the outer-most term, connect the **OR** term transistors in parallel, and the **AND** term transistors in series
- For the p-transistor tree, working from the inner-most bracket to the outer-most term, connect the **OR** term transistors in series, and the **AND** term transistors in parallel

# Pseudo-nMOS Logic Gates




W&H p335-336



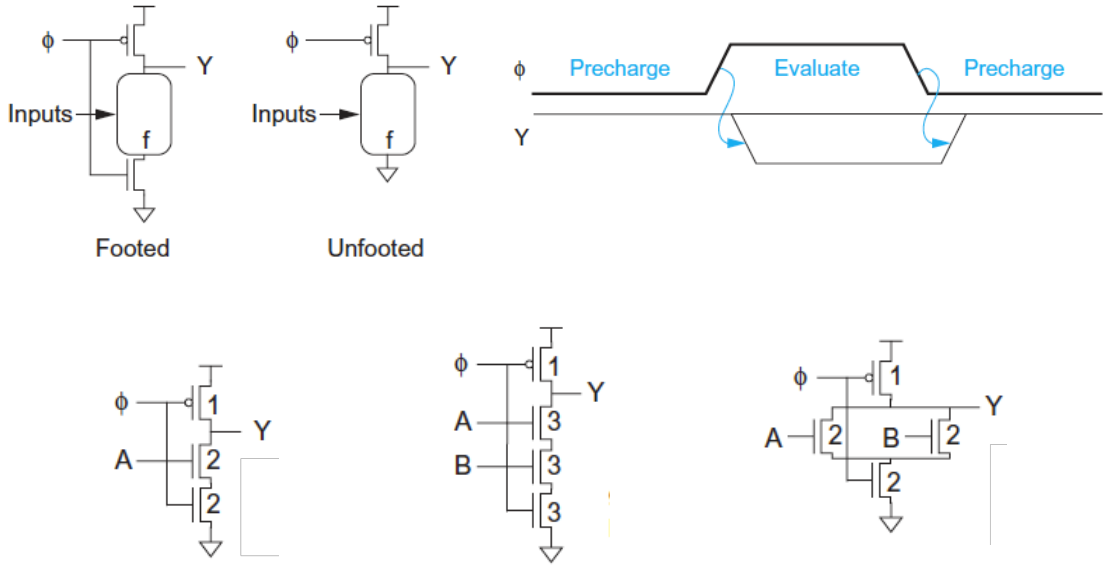
The slide shows several pseudo-nMOS logic gates. The pulldown network is like that of an ordinary static gate in nMOS, but the pullup network has been replaced with a single pMOS transistor that is grounded so it is always ON. The pMOS transistor widths are selected to be about 1/4 the strength (i.e., 1/2 the effective width) of the nMOS pulldown network as a compromise between noise margin and speed. (Assuming pMOS mobility is half that of nMOS.)

Pseudo-nMOS gates is smaller in area (no pMOS network except the pullup, always-ON, transistor. It can be faster because the input capacitance is reduced by replacing the pullup network with one always-ON pMOS transistor.

Unfortunately, it suffers many disadvantages including:

- Static power dissipation through the always-ON pMOS transistor.
- Poorer noise margin due to ratio logic and output high level is never at VDD.
- Correct operation require careful choice of ratio between pull-up and pull-down strength.
- Bad with low voltages (therefore not useful in latest technology).
- Slow rising transition.

## Dynamic Logic Gates




W&H p339-341



To avoid problem with pseudo-nMOS logic, we replace the always-ON pullup pMOS with a clocked pMOS transistor driven by a clock signal. This is called a **dynamic logic gate**.

- When the clock = 0, the output node is precharged to VDD. This is the precharge phase of the clock.
- When the clock = 1, the nMOS network evaluate the logic function. This is the evaluation phase of the clock.

If the output Y is high, the precharged node never discharge. Y is not driven. Therefore this logic gate is dynamic in nature. It cannot retain its output forever. As a result, the gate will not work properly unless there is constant clock precharging and evaluating.

The **advantage** of dynamic logic are:

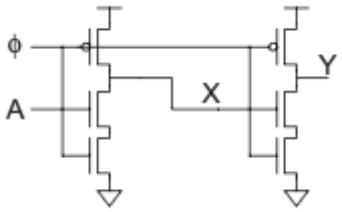
- Faster due to lower gate capacitance.
- No static power (unlike pseudo-nMOS).

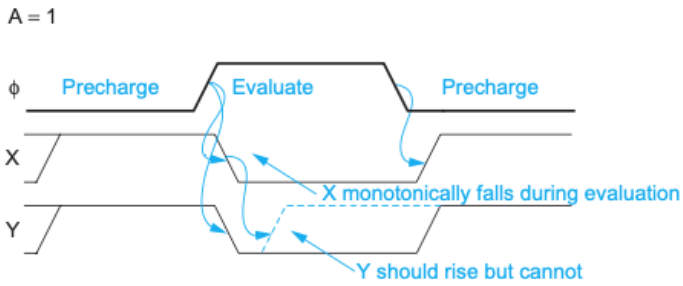
The **disadvantages** of dynamic logic are:

- Dissipate more dynamic power.
- Sensitive to noise during evaluation phase.
- Suffer from “monotonicity” problem.

Nowadays, dynamic logic is still used, but only for speed critical operations where robustness is not an issue. Any dynamic logic circuit must be designed with utmost care.

# Problem with Dynamic Logic Gates




W&H p341



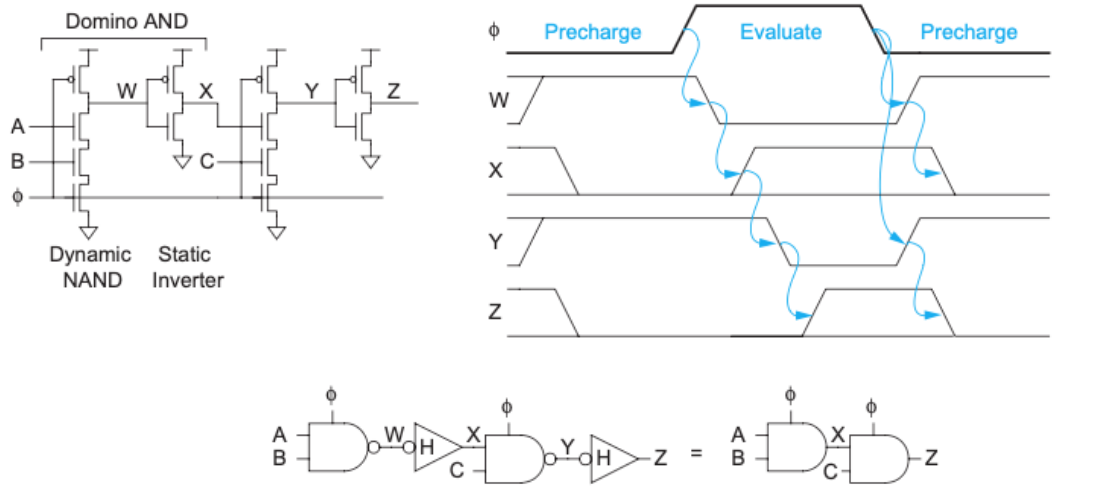
The problem of dynamic logic gate is that it does not work properly unless the Inputs are **monotonically rising**. That is,

- Inputs start LOW and remain LOW, or
- Inputs start LOW and rise HIGH, or
- Inputs start HIGH and remain HIGH.

However, inputs **MUST NOT** start HIGH and fall LOW during evaluation.

Consider the two dynamic inverter shown in the slide. If input A is initially HIGH, but goes low in evaluation phase, node X is already discharge to VSS before A does low. X will never go back to '1' again! The inverter does not work like an inverter anymore!

# Domino Logic




W&H p341-342



The monotonicity problem can be solved by placing a static CMOS inverter between dynamic gates. This converts the monotonically falling output into a monotonically rising signal suitable for the next gate.

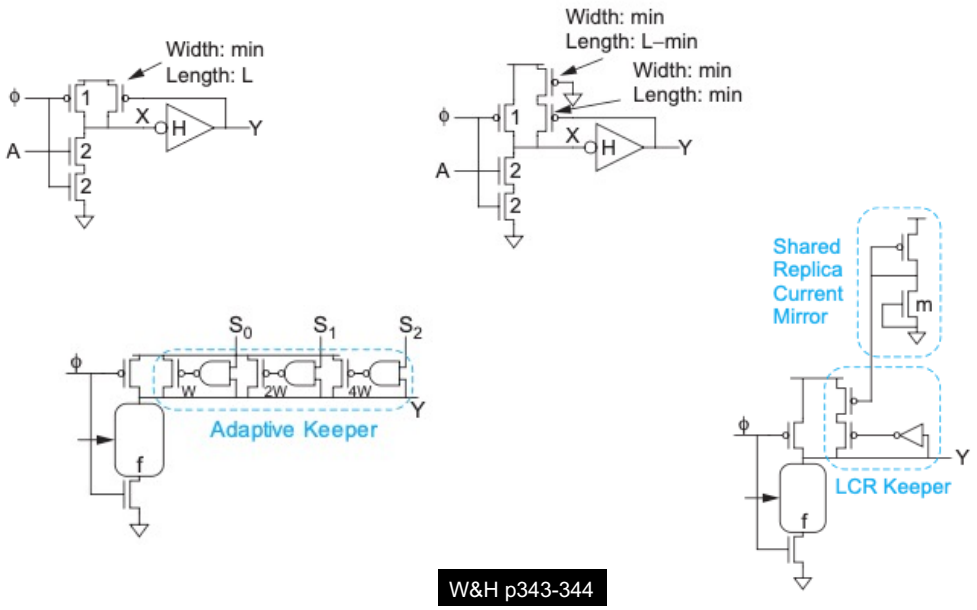
The dynamic-static pair together is called a domino gate because precharge resembles setting up a chain of dominos and evaluation causes the gates to fire like dominos tipping over, each triggering the next.

A single clock can be used to precharge and evaluate all the logic gates within the chain. The dynamic output is monotonically falling during evaluation, so the static inverter output is monotonically rising. Therefore, the static inverter is usually designed such that the output rises faster than it falls. This is called a HI-skew gate because it skews towards rising output instead of having balance rise and fall times.

Note that precharge occurs in parallel, but evaluation occurs sequentially.

The static inverter can be replace with more complex CMOS gates such as NAND or NOR. This mixture of dynamic and static logic is called compound domino.

# Keepers for Dynamic Logic




W&H p343-344



Dynamic circuits suffer from charge leakage on the dynamic node. If a dynamic node is precharged high and then left floating, the voltage on the dynamic node will drift over time due to leakages. The time constants tend to be in the millisecond to nanosecond range, depending on process and temperature.

Moreover, dynamic circuits have poor input noise margins. Any noise to the input could incorrectly discharge the output. Both leakage and noise margin problems can be addressed by adding a keeper circuit.

The keeper is a weak transistor that holds the output at the correct level when it would otherwise float. Consider the circuit on the top left. When the dynamic node X is high, the output Y is low and the keeper is ON to prevent X from floating. When X falls, the keeper initially opposes the transition so it must be much weaker than the pulldown network. Eventually Y rises, turning the keeper OFF and avoiding static power dissipation.

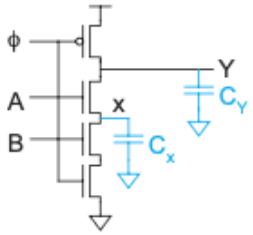
The keeper must be strong (i.e., wide) enough to compensate for any leakage current drawn when the output is floating and the pulldown stack is OFF.

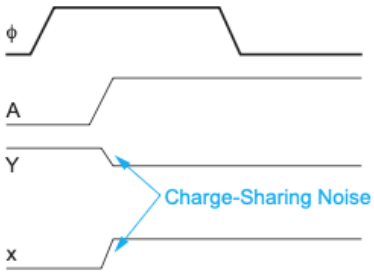
For small dynamic gates, the keeper must be weaker than a minimum-sized transistor. This is achieved by increasing the keeper length.

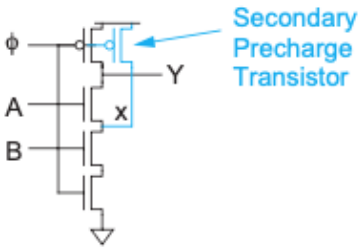
However, long keeper transistors increase the capacitive load on the output Y. This can be avoided by splitting the keeper, as shown in the top right circuit.

Two other variants are shown here. One has digitally programmable keeper current (bottom left). The second uses a current mirror. The size m is adjusted to match the leakage current and made to track with both process variations and temperature.

# Charge Sharing in Dynamic logic




W&H p345



Dynamic gates are subject to problems with charge sharing. For example, consider the 2-input dynamic NAND gate shown above. Suppose the output Y is precharged to VDD and inputs A and B are low. Also suppose that the intermediate node x had a low value from a previous cycle. During evaluation, input A rises, but input B remains low so the output Y should remain high. However, charge is shared between Cx and CY. This behaves as a capacitive voltage divider and the voltages equalize at:

$$V_x = V_Y = \frac{C_Y}{C_x + C_Y} V_{DD}$$

Charge sharing can be overcome by precharging some or all of the internal nodes with secondary precharge transistors (right circuit). It is often sufficient to precharge every other node in a tall stack.